4	The variation with potential difference *V of* current I in a resistor X is shown in Fig. 4.1.		

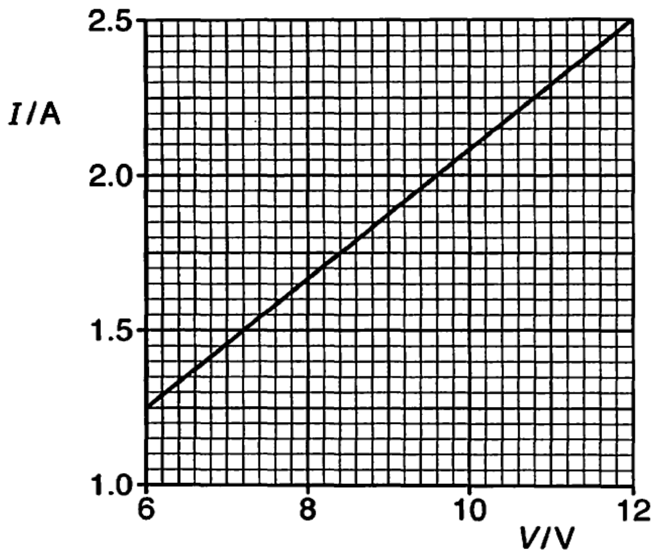


Fig. 4.1

	(a)	Use Fig. 4.1 to show that the resistance of X remains constant.	
			[2]

Method:

- *calculate the RATIO of the V-coordinate to the I-coordinate.*
- *Compare the ratios from the different points. Show that the ratios are constant.*

At (6.0 V, 1.25 A), $R = \frac{V}{I} = \frac{6.0}{1.25} = 4.8\ \Omega$ **1**

At (12 V, 2.5 A), $R = \frac{V}{I} = \frac{12}{2.5} = 4.8\ \Omega$

Since the ratio $\frac{V}{I}$ remains constant at 4.8 Ω, resistance of X remains constant. **1**

Method 2:

- Calculate gradient with either 2 points that are far apart on the graph OR at least 3 different points along the graph. **1**
- Use the gradient and straight line equation to calculate y-intercept and show that it is equal to zero. **1**

(b) In an attempt to obtain the graph of Fig. 4.1 for resistor X, a student sets up a circuit as shown in Fig. 4.2.

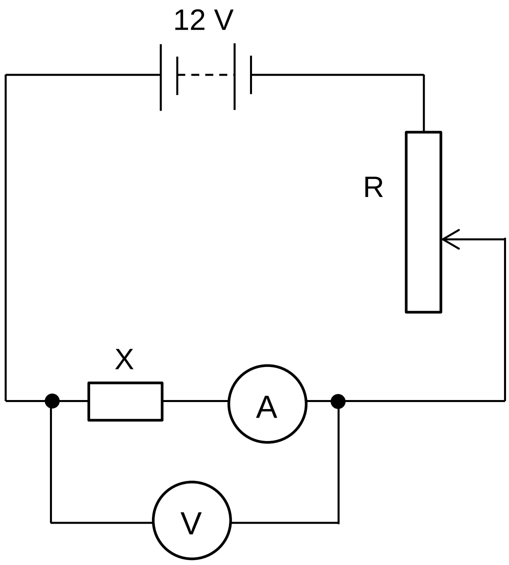


Fig. 4.2

The battery has an e.m.f. of 12 V and negligible internal resistance. The resistor R has a resistance that can be varied between 0.0 Ω and 2.0 Ω. The voltmeter and ammeter are both ideal.

State and explain why the circuit shown in Fig. 4.2 is inappropriate for obtaining data from V = 6 V to V = 12 V for the graph in Fig. 4.1.

....

....

....

....

.... [2]

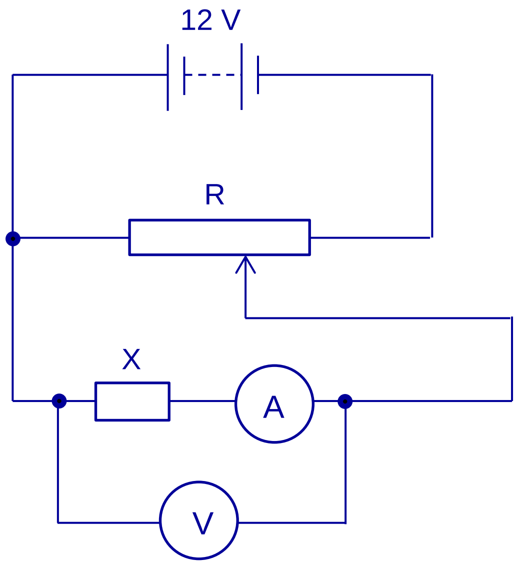
When R is set to its minimum of 0 Ω, a maximum potential difference (p.d.) of 12 V can be applied across X. Hence voltmeter and ammeter readings up to V = 12 V are obtainable. **1**

However, when R is set to its maximum of 2.0 Ω, since X has resistance of 4.8 Ω which is greater than 2.0 Ω, the p.d. across X will be greater than 6 V ($\frac{4.8}{4.8+2.0} x 12 = 8.5\ V$). It would thus be impossible to obtain voltmeter and ammeter readings in the range of V = 6 V to V = 8.5 V in Fig. 4.1. **1**

(c) In the space below, draw a circuit diagram using the same components as shown in Fig. 4.2 from which the graph of Fig. 4.1 may be determined.

[1]

Solution:



1

(d) Calculate the difference in the power dissipated in resistor X when *V* is increased from 7.2 V to 9.6 V.

difference in power = ... W [2]

Solution:

$Difference\ in\ power = I_1V_1 - I_2V_2$ **1**

$= (2.0)(9.6) - (1.5)(7.2)$

$= 8.4\ W$ **1**

(e) Suppose the battery has an internal resistance of 0.50 Ω, and R in Fig. 4.2 is adjusted to 0 Ω.

Calculate the terminal potential difference across the battery.

terminal potential difference = ... V [2]

By potential divider principle,

$terminal\ p.d. = p.d.across\ X = \frac{4.8}{4.8 + 0.5}\ x\ 12\ V$ **1**

= 10.868 V = **11 V** **1**

5	**(a)**	When white light is incident on a single slit, a diffraction pattern is formed on a screen.